

20/3/26

Exp-Ta

IPC [shared memory]

1) sender:

Aim:

To implement inter process communication (IPC) using shared memory in C, where one process writes data into a shared memory segment.

Algorithm:

- the start program
- define a unique key for shared memory
- create a shared memory to the process using `shmget()`
- attach the shared memory using `shmat()`.
- prompt the user to enter message
- read the input string from user
- store the input string directly into shared mem.
- display message stored
- end program

Code:

```
#include <stdio.h>
#include <sys/types.h>
#include <sys/shm.h>
#include <string.h>

int main() {
    key_t ky = 1234
```



```
int schmid;  
schmid = shmget(key, 1024, 0666, IPC_CREAT);  
char *str = (char *) shmctl(schmid, 0);  
printf("Enter message");  
scanf("%[^\n]", str);  
printf("Data written to shared memory: %s\n", str);  
return 0;
```

}

Output:

Enter message: Hello

Data written to shared memory: Hello

Result:

The program successfully creates a shared memory segment and writes user input to it.

2. Receiver

Aim:

To read data from a shared memory segment using IPC in C, where receiver access and display message.

Algorithm:

- start the program
- define the same unique key
- access the existing shared memory
- attach the shared memory to produce
- read the data
- display the message on screen
- detach the shared memory using shmat
- end the program

Code:

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
int main() {
    key_t key = 1234;
    int shmid;
    shmid = shmget(key, 1024, 0666);
    char *str = (char *) shmat(shmid, 0, 0);
    printf("Data read %s\n", str);
    shmat(str);
    return 0; }
```


Output :

Data read from shared memory: Hello World

Data written to shared memory: Hello World

Result :

The program successfully accessed the executable memory.

